Chapter 3: Number Theory

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3.1. Division

Definition.

Let a and b be integers, then b is said to be *divisible by* a, or b is a *multiple* of a, or a is a *divisor* of b, or a *divides* b, if there exists an integer k such that $b = ka$.
If a devides b, we write $a \mid b$, otherwise, we write $a \nmid b$.

Example. $3 \nmid 7$ because $3 \times 2 = 6 < 7 < 3 \times 3 = 9$
Example. $3 \mid 12$ because $3 \times 4 = 12$
The following statements are equivalent

- $a \mid b$,
- a divides b,
- a is a *divisor* of b,
- a is a *factor* of b,
- b is *divisible* by a,
- b is a *multiple* of a.

Theorem

Let a, b, c be integers, $a \neq 0$. Then:

(i) If $a \mid b$ and $a \mid c$ then $a \mid (b + c)$;

(ii) If $a \mid b$ then $a \mid bc$ for any integer c;

(iii) If $a \mid b$ and $b \mid c$ then $a \mid c$.

Proof: Homework.

3.1. Division Algorithm

Division Algorithm. Let a be an integer and d a positive integer. Then there exist unique integers q and r such that

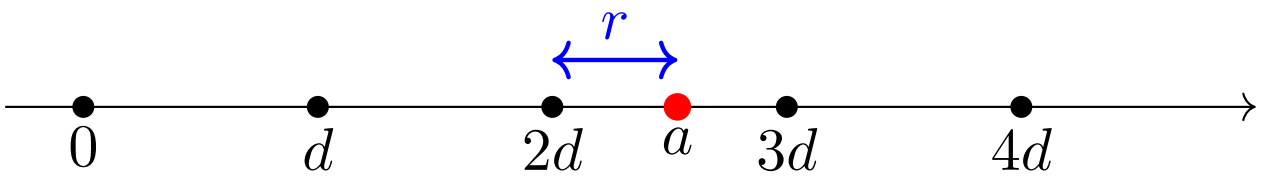
$$a = dq + r, \quad 0 \leq r < d.$$

- a: *dividend*
- d: *divisor*
- q: *quotient*
- r: *remainder*

We also write:

$$q = a \text{ div } d, \quad r = a \bmod d.$$

Example: $a = d \cdot 2 + r$



Division Examples

Example. Find the quotient and the remainder in the division of 101 by 11.

Solution. We have

$$101 = 11 \times 9 + 2$$

Hence $9 = 101 \operatorname{div} 11$ and $2 = 101 \bmod 11$.

Example. Find the quotient and the remainder in the division of -11 by 3.

Solution. We have

$$-11 = 3 \times (-4) + 1$$

Hence $-4 = -11 \operatorname{div} 3$ and $1 = -11 \bmod 3$.

Note. We may write

$$-11 = 3 \times (-3) - 2$$

However -2 is not the remainder r because r always satisfies $0 \leq r < d$.

What is a prime?

Definition

A **prime** is an integer $p > 1$ divisible only by 1 and p.

Composite

An integer not equal to $-1, 0, 1$ that is not prime is **composite**.

Examples: $2, 3, 5, 7, 11$ are prime; $9 = 3^2$, $21 = 3 \cdot 7$ are composite.

Euclid's Lemma

Lemma

If p is prime and $p \mid a_1 a_2 \cdots a_k$, then $p \mid a_i$ for some i.

Idea of proof: For $k = 2$, if $p \nmid a$ then $\gcd(p, a) = 1$. From $p \mid ab$ we get $p \mid b$. Induct on k.
Quick check: $5 \mid 35 = 5 \cdot 7 \Rightarrow 5 \mid 5$ or $5 \mid 7$.

Fundamental Theorem of Arithmetic

Statement

Every integer greater than 1 can be written uniquely as a prime or as the product of two or more primes where the prime factors are written in order of nondecreasing size.

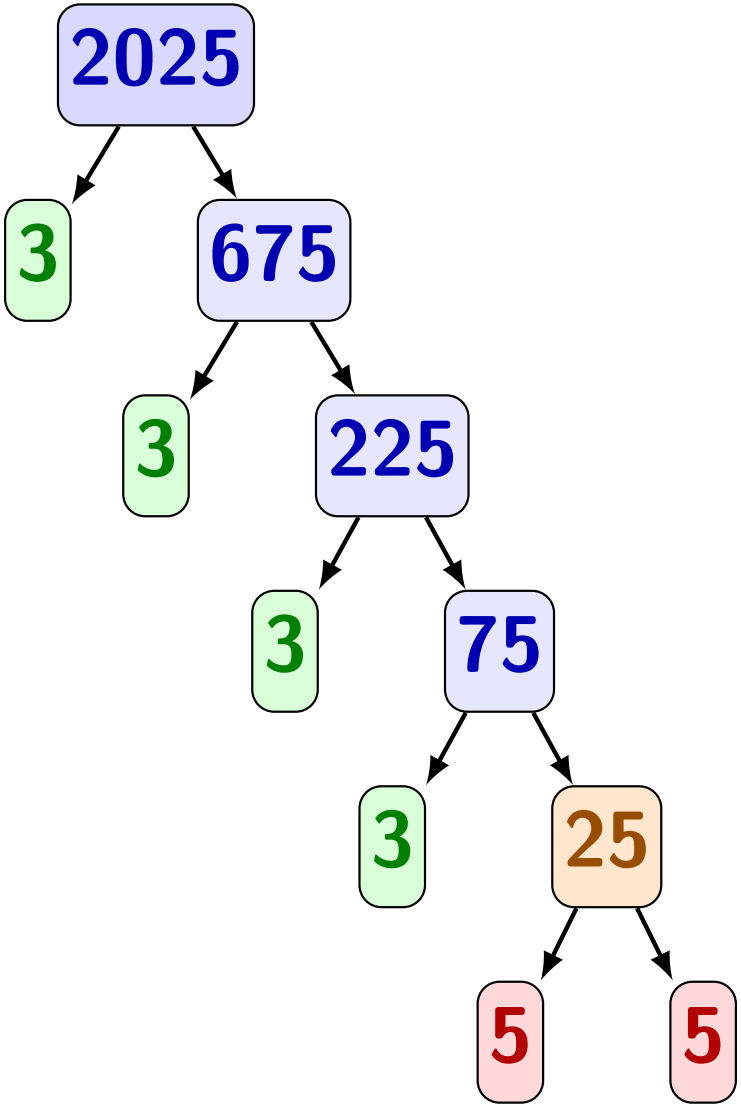
Examples:

$$12 = 2^2 \cdot 3, \quad 20 = 2^2 \cdot 5, \quad 30 = 2 \cdot 3 \cdot 5, \quad 504 = 2^3 \cdot 3^2 \cdot 7.$$

Theorem

If n is a composite integer, then n has a prime divisor less than or equal to $\sqrt{n}$.

Factor 2025



$$2025 = 3^4 \times 5^2$$

Prime Mysteries

Conjectured Inefficiency of Factoring

Given $n = pq$ with large primes p, q, no known *polynomial-time* algorithm is guaranteed to recover p and q from n (measured in input length).

This fuels public-key cryptography (e.g., RSA).

Goldbach's Conjecture

Conjecture

Every even integer > 2 is the sum of two primes.

Examples: $4 = 2+2,\ 6 = 3+3,\ 8 = 3+5,\ 28 = 11+17,\ 100 = 47+53.$
We know: Every even number is a sum of at most 6 primes (weak results).

Prime Number Theorem (PNT)

Let $\pi(n)$ be the number of primes $\leq n$.

Theorem

$$\lim_{n\to\infty} \frac{\pi(n)}{n/\ln n} = 1.$$

Meaning: $\pi(n) \sim n/\ln n$.
Rule of thumb: density near n is about $1/\ln n \Rightarrow$ roughly one prime in every $\ln n$ integers.

Chebyshev's Lower Bound

For $n > 1$

$$\pi(n) > \frac{n}{3 \ln n}.$$

Example: $\pi(10^6) > \dfrac{10^6}{3 \ln 10^6} \approx 24{,}119$ (actual: $78{,}498$).

A Prime for Google

In 2004, a billboard asked for the *first 10-digit prime* appearing among consecutive digits of e.

Back-of-the-envelope

Among 10-digit numbers, the fraction that are prime is $\approx 1/\ln(10^{10}) = 1/(10\ln 10) \approx 1/23$.

Exercise: Roughly how many 10-digit primes are there?

A Prime for Google

In late 2004 a billboard appeared in various locations around the country:

$$\left\{ \begin{array}{l} \text{first 10-digit prime found} \\ \text{in consecutive digits of } e \end{array} \right\}\text{.com}$$

Substituting the correct number for the expression in curly-braces produced the URL for a Google employment page. The idea was that Google was interested in hiring the sort of people that could and would solve such a problem.

$$e = 2.718281828459045235360287471352662497757247093699959574966$$

$$967627724076630353547594571382178525166427\mathbf{427466391}9320030$$

Worked Examples

1. Decide prime/composite: 97, 221, 257.
2. Find the largest prime dividing $\binom{100}{50}$.
3. Estimate $\pi(10^7)$ using PNT; compare with Chebyshev's bound.

Summary

- Primes and unique factorization underpin arithmetic.
- Hardness of factoring motivates modern cryptography.
- Distribution: $\pi(n) \sim n/\ln n$; density $\approx 1/\ln n$.
- Open problem: Goldbach's conjecture.

The Euclidean Algorithm — Definitions

Greatest common divisor. Let a and b be *nonzero* integers. The largest positive integer d that divides both a and b is called the greatest common divisor of a and b, denoted $\gcd(a, b)$.
Two integers a, b are *relatively prime* if $\gcd(a, b) = 1$.

Least common multiple (LCM). For nonzero integers a, b, the least common multiple $\operatorname{lcm}(a, b)$ is the least positive integer divisible by both a and b.
For $a, b > 0$, we have

$$\gcd(a, b) \cdot \operatorname{lcm}(a, b) = ab$$

Example 1

$24 = 2^3 \cdot 3$ and $60 = 2^2 \cdot 3 \cdot 5$

$gcd(24, 60) = 12$

$lcm(24, 60) = 2^3 \cdot 3 \cdot 5 = 120$ or $lcm(24, 60) = \frac{24 \cdot 60}{gcd(24,60)} = 120$

Example 2. The integers 17 and 22 have no positive common divisors other than 1, so that gcd(17, 22) $= 1$. Moreover, 17 and 22 are relatively prime.

Lemma

Let $a = bq + r$, where a, b, q, and r are integers. Then $gcd(a, b) = gcd(b, r)$.

Example: Compute $\gcd(287, 91)$
First divide 287 by 91: $\quad 287 = 91 \cdot 3 + 14.$

Then we have

$$\gcd(287, 91) = \gcd(91, 14).$$

Next divide 91 by 14: $\quad 91 = 14 \cdot 6 + 7.$
We get $\gcd(91, 14) = \gcd(14, 7)$.
Continue: $\quad 14 = 7 \cdot 2 + 0$ so $\gcd(14, 7) = 7$.

Therefore $\boxed{\gcd(287, 91) = 7}$.

The Euclidean Algorithm (general form)

Let a, b be positive integers with $a \geq b$. Set $r_0 = a,\ r_1 = b$. Apply the division algorithm successively:

$$\begin{aligned}
r_0 &= r_1 q_1 + r_2, & 0 \leq r_2 < r_1,\\
r_1 &= r_2 q_2 + r_3, & 0 \leq r_3 < r_2,\\
&\vdots\\
r_{n-2} &= r_{n-1} q_{n-1} + r_n, & 0 \leq r_n < r_{n-1},\\
r_{n-1} &= r_n q_n.
\end{aligned}$$

Finally we get $\boxed{\gcd(a, b) = r_n}$.

Example: $\gcd(662, 414)$ via Euclidean Algorithm

$$\begin{aligned} 662 &= 414 \cdot 1 + 248, \\ 414 &= 248 \cdot 1 + 166, \\ 248 &= 166 \cdot 1 + 82, \\ 166 &= 82 \cdot 2 + 2, \\ 82 &= 2 \cdot 41 + 0. \end{aligned}$$

Hence $\boxed{\gcd(662, 414) = 2}$.

Example: Prove that $gcd(3a, 5a+1) = gcd(a-1, 3)$ for any integer $a > 1$.
Since $5a+1 = (3a) \cdot 1 + (2a+1)$, we have:

$$gcd(3a, 5a+1) = gcd(3a, 2a+1)$$

Since $3a = (2a+1) \cdot 1 + (a-1)$, we have

$$gcd(3a, 2a+1) = gcd(2a+1, a-1)$$

Similarly, we have $2a+1 = (a-1) \cdot 2 + 3$ nên

$$gcd(2a+1, a-1) = gcd(a-1, 3).$$

This implies that $gcd(3a, 5a+1) = gcd(a-1, 3)$.

Theorem

Bezout's theorem If a and b are positive integers, then there exist integers s and t such that $gcd(a, b) = sa + tb$.

Definition

If a and b are positive integers, then integers s and t such that $gcd(a, b) = sa + tb$ are called Bézout coefficients of a and b.

Find Bezout coefficients for a and b

Example

Find integers s, nt such that $662s + 414t = \gcd(662, 414)$.

Euclidean steps

$$\begin{aligned} 662 &= 414 \cdot 1 + 248, \\ 414 &= 248 \cdot 1 + 166, \\ 248 &= 166 \cdot 1 + 82, \\ 166 &= 82 \cdot 2 + 2, \\ 82 &= 2 \cdot 41 + 0. \end{aligned}$$

Therefore $\gcd(662, 414) = 2$.

Back substitution to get s, t

$$\begin{aligned} 2 &= 166 - 82 \cdot 2, \\ &= 166 - (248 - 166 \cdot 1) \cdot 2 = 248 \cdot (-2) \\ &= 248 \cdot (-2) + (414 - 248 \cdot 1) \cdot 3 \\ &= 414 \cdot 3 + 248 \cdot (-5), \\ &= 414 \cdot 3 + (662 - 414 \cdot 1) \cdot (-5) \\ &= 662 \cdot (-5) + 414 \cdot 8. \end{aligned}$$

Hence $\boxed{s = -5,\ t = 8}$ and $662(-5) + 414(8) = 2$.

Definition and basic notation

Definition. Let m be a positive integer and $a, b \in \mathbb{Z}$. We say that a *is congruent to* b *modulo* m if $m \mid (a - b)$. We write

$$a \equiv b \pmod{m}.$$

If a is not congruent to b modulo m we write $a \not\equiv b \pmod{m}$.

Theorem. Let $a, b \in \mathbb{Z}$ and $m \in \mathbb{Z}_{>0}$. Then

$$a \equiv b \pmod{m} \iff a \bmod m = b \bmod m.$$

Proof of the theorem

Assume $a \equiv b \pmod{m}$. By the division algorithm write

$$a = mq + r, \quad 0 \le r < m \qquad \text{and} \qquad b = mq' + r', \quad 0 \le r' < m.$$

Then $m \mid (a - b)$ iff m divides

$$a - b = (mq + r) - (mq' + r') = m(q - q') + (r - r').$$

Since $0 \le |r - r'| < m$, the only multiple of m in this range is 0; hence $r - r' = 0$, i.e. $r = r'$. Therefore $a \bmod m = r = b \bmod m$.

The converse is immediate: if $a \bmod m = b \bmod m$ then $r = r'$ in the above expressions, hence $a - b = m(q - q')$ and $m \mid (a - b)$, i.e. $a \equiv b \pmod{m}$.

Example

Are the following congruences true?

$$17 \equiv 5 \pmod 6 \qquad \text{and} \qquad 24 \equiv 14 \pmod 6 \ ?$$

Solution 1.

- Since $17 - 5 = 12$ is divisible by 6, we have $17 \equiv 5 \pmod 6$.
- Since $24 - 14 = 10$ is not divisible by 6, we conclude

$$24 \not\equiv 14 \pmod 6.$$

Solution 2:

- We have $17 \bmod 6 = 5$ and $41 \bmod 6 = 5$ so $17 \equiv 41 \pmod 6$
- We have $24 \bmod 6 = 0$ and $14 \bmod 6 = 2$ so $17 \not\equiv 41 \pmod 6$.

Theorem

Theorem

Let a, b be integers and let m be a positive integer. Then

$$a \equiv b \pmod{m} \iff \exists\, k \in \mathbb{Z} \text{ such that } a = b + k\,m.$$

Proof

Assume $a \equiv b \pmod{m}$. Prove that $a = b + k\,m$ for some $k \in \mathbb{Z}$.

By the division algorithm, for some integers q, q', r, r' with $0 \le r, r' < m$,

$$a = mq + r, \qquad b = mq' + r'.$$

Since $a \equiv b \pmod{m}$ we have $m \mid (a - b)$. But

$$a - b = (mq + r) - (mq' + r') = m(q - q') + (r - r').$$

Because $0 \le |r - r'| < m$, the only multiple of m in this interval is 0, thus $r - r' = 0$ (i.e. $r = r'$). Hence

$$a - b = m(q - q') \quad \Rightarrow \quad a = b + m(q - q').$$

So $k = q - q' \in \mathbb{Z}$ works.

Conversely, if $a = b + k\,m$ for some $k \in \mathbb{Z}$, then $a - b = km$ and therefore $m \mid (a - b)$, i.e. $a \equiv b \pmod{m}$.

Theorem

Let m be a positive integer. If $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$, then

1. $a \pm c \equiv b \pm d \pmod{m}$,
2. $ac \equiv bd \pmod{m}$,
3. $a^k \equiv b^k \pmod{m} \quad \forall\, k \in \mathbb{N}$.

Example

Since $7 \equiv 2 \pmod 5$ and $11 \equiv 1 \pmod 5$, we have

$$\begin{aligned} 18 &= 7 + 11 \equiv 2 + 1 = 3 \pmod 5, \\ 77 &= 7 \cdot 11 \equiv 2 \cdot 1 = 2 \pmod 5, \\ -4 &= 7 - 11 \equiv 2 - 1 = 1 \pmod 5. \end{aligned}$$

More Illustrations

- If $23 \equiv 2 \pmod 7$ and $15 \equiv 1 \pmod 7$ then

$$23 + 15 \equiv 2 + 1 = 3 \pmod 7, \quad 23 \cdot 15 \equiv 2 \cdot 1 = 2 \pmod 7.$$

- $32 \equiv 2 \pmod 5 \Rightarrow 32^4 \equiv 2^4 = 16 \equiv 1 \pmod 5$.
- $a \equiv b \pmod 9 \Rightarrow 5a - 3 \equiv 5b - 3 \pmod 9$ (linear combination).

Definition and Examples

Definition. For a given integer b, if there exists an integer c such that

$$bc \equiv 1 \pmod{N},$$

then we say that b is *invertible modulo* N and call c a (multiplicative) *inverse* of b modulo N.

Examples.

$$2 \cdot 3 \equiv 1 \pmod{5}, \qquad 2 \cdot 8 \equiv 1 \pmod{15}.$$

Hence the inverse of 2 modulo 5 is 3, and the inverse of 2 modulo 15 is 8.

Proposition and Method

Proposition. Let $b, N > 1$ be integers. Then b is invertible modulo N

$$\iff \quad \gcd(b, N) = 1.$$

How to find the inverse of b modulo N

1. Use the *extended Euclidean algorithm* to find integers x, y with

$$bx + Ny = 1.$$

2. Reduce this identity modulo N to get $bx \equiv 1 \pmod{N}$. Conclude that x is an inverse of b modulo N (take $x \bmod N$ in $0, \ldots, N-1$).

Exercise

Let $\mathbb{Z}_{26} = \{0, 1, 2, \ldots, 25\}$.

(a) List all positive integers in $\mathbb{Z}_{26}$ that are invertible modulo 26.

(b) Let $a, b \in \mathbb{Z}$. Consider the map

$$f : \mathbb{Z}_{26} \longrightarrow \mathbb{Z}_{26}, \qquad f(x) \equiv ax + b \pmod{26}.$$

(i) Prove that $f(x) = 3x$ is bijective and find its inverse function.
(ii) Prove that $f(x) = 3x + b$ is bijective for any b.
(iii) Find all values of a for which f is bijective.

Theorem (Chinese Remainder Theorem)

Let $m_1, \ldots, m_k$ be pairwise coprime and set $M = \prod_{i=1}^{k} m_i$. For any residues $a_1, \ldots, a_k$, the system

$$x \equiv a_i \pmod{m_i} \quad (i = 1, \ldots, k)$$

has a *unique* solution modulo M.

Uniqueness: If x, y both satisfy all congruences, then $m_i \mid (x - y)$ for all i, hence $M \mid (x - y)$ and $x \equiv y \pmod{M}$.

Explicit Construction & Diagram

Construction. Set $M = \prod_{i=1}^{k} m_i$ and $M_i = \dfrac{M}{m_i}$. Choose N_i such that

$$M_i N_i \equiv 1 \pmod{m_i}.$$

Then a solution is

$$x \equiv \sum_{i=1}^{k} a_i \, M_i \, N_i \pmod{M}.$$

(Proof idea: reduce the sum modulo each m_i and use $M_i N_i \equiv 1 \pmod{m_i}$.)

$$\begin{tikzcd}
 & \boxed{\mathbb{Z}/M\mathbb{Z}} \arrow[dl, "\bmod m_1"'] \arrow[dr, "\bmod m_k"] & \\
\boxed{\mathbb{Z}/m_1\mathbb{Z}} & \cdots & \boxed{\mathbb{Z}/m_k\mathbb{Z}}
\end{tikzcd}$$

(Ring isomorphism)

$$\mathbb{Z}/M\mathbb{Z} \cong \prod_{i=1}^{k} \mathbb{Z}/m_i\mathbb{Z}$$

Example 1

Solve the following system:

$$\begin{cases} x \equiv 2 \pmod 3 \\ x \equiv 3 \pmod 5 \end{cases}$$

Solution. Here $m_1 = 3, m_2 = 5, M = 15$
Step 1. Compute $M_1 = \frac{M}{m_1} = 5, \qquad M_2 = \frac{M}{m_2} = 3.$
Step 2. Find N_i such that $M_i N_i \equiv 1 \pmod p$:

$$5N_1 \equiv 1 \pmod 3 \implies N_1 = 2.$$

$$3N_2 \equiv 1 \pmod 5 \implies N_2 = 2.$$

Step 3. Compute the general solution

$$x \equiv a_1 M_1 N_1 + a_2 M_2 N_2 \pmod M.$$

We have $\quad x \equiv 2 \cdot 5 \cdot 2 + 3 \cdot 3 \cdot 2 \pmod{15}$ and thus $x \equiv 38 \pmod{15}$.
The unique solution modulo 15 is:

$$\boxed{x \equiv 8 \pmod{15}}$$

Example 2

Solve:

$$x \equiv 2 \pmod 3, \qquad x \equiv 3 \pmod 4, \qquad x \equiv 1 \pmod 5.$$

Compute $M = 3 \cdot 4 \cdot 5 = 60$.

$M_1 = \frac{60}{3} = 20, \quad M_2 = \frac{60}{4} = 15, \quad M_3 = \frac{60}{5} = 12.$
Find inverses N_i:
$20N_1 \equiv 1 \ (\text{mod} 3) \Rightarrow N_1 = 2, \quad 15N_2 \equiv 1 \ (\text{mod} 4) \Rightarrow N_2 = 3,$
$12N_3 \equiv 1 \ (\text{mod} 5) \Rightarrow N_3 = 3.$
Assemble:

$$\begin{aligned} x &\equiv 2 \cdot 20 \cdot 2 + 3 \cdot 15 \cdot 3 + 1 \cdot 12 \cdot 3 \pmod{60} \\ &= 80 + 135 + 36 = 251 \equiv \boxed{11} \pmod{60}. \end{aligned}$$

Check: $11 \bmod 3 = 2$, $11 \bmod 4 = 3$, $11 \bmod 5 = 1$.

Mini exercise

Solve:

$$\begin{cases} x \equiv 1 \pmod{2}, \\ x \equiv 2 \pmod{3}, \\ x \equiv 3 \pmod{5}. \end{cases} \pmod{30}$$

Fermat's Little Theorem and Example

Theorem

If p is a prime and a is an integer not divisible by p, then

$$a^{p-1} \equiv 1 \pmod{p}.$$

Equivalently, for any integer a,

$$a^p \equiv a \pmod{p}.$$

Example: Take $p = 7$, $a = 3$. Since $\gcd(3, 7) = 1$:

$$3^{7-1} = 3^6 = 729.$$

Now divide by 7:

$$729 = 7 \cdot 104 + 1.$$

So

$$3^6 \equiv 1 \pmod{7}.$$

Fermat's Little Theorem and Example

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$$729 = 7 \cdot 104 + 1.$$

So

$$3^6 \equiv 1 \pmod{7}.$$

Euler's Theorem

Definition

For any positive integer n, $\varphi(n)$ is the number of positive integers less than n that are coprime to n.
$\varphi(n)$ is called **Euler's totient function**.

Theorem (Euler's theorem)

Let n be a positive integer and a be an integer such that $\gcd(a, n) = 1$. Then:

$$a^{\varphi(n)} \equiv 1 \pmod{n}$$

Example of Euler's Theorem

Example: Verify Euler's theorem for $a = 3$ and $n = 10$.

1. Check $\gcd(a, n)$:
$$\gcd(3, 10) = 1 \quad \checkmark$$
2. Compute Euler's totient function:
$$n = 10 = 2 \cdot 5 \quad \Rightarrow \quad \varphi(10) = 10\left(1 - \frac{1}{2}\right)\left(1 - \frac{1}{5}\right) = 4$$
3. Verify theorem:
$$3^{\varphi(10)} = 3^4 = 81 \equiv 1 \pmod{10}$$

Conclusion: $3^4 \equiv 1 \pmod{10}$ $\checkmark$

Proof of Euler's theorem

Proof.

Let $R = \{r_1, \ldots, r_{\varphi(n)}\}$ be a reduced residue system modulo n. Since $\gcd(a, n) = 1$, the set $\{ar_1, \ldots, ar_{\varphi(n)}\}$ is also a reduced residue system modulo n, hence a permutation of R. Therefore

$$(ar_1) \cdots (ar_{\varphi(n)}) \equiv r_1 \cdots r_{\varphi(n)} \pmod{n}.$$

Factoring gives

$$a^{\varphi(n)} \cdot (r_1 \cdots r_{\varphi(n)}) \equiv r_1 \cdots r_{\varphi(n)} \pmod{n}.$$

Because each r_i is coprime to n, their product is invertible modulo n, so we may cancel to obtain $a^{\varphi(n)} \equiv 1 \pmod{n}$. □

Application of Congruences

- Hash function
- Pseudorandom number

Hashing function

- Hashing functions should be easily evaluated.
- A hashing function is noth always one-to-one.
- We say that a *collision* occurs if there exist x, y such that $H(x) = H(y)$.
- A hashing function is called *collision-resistent* if it is hard to find two different inputs x_1, x_2 such that $H(x_1) = H(x_2)$.
- The outputs of a hashing function are usually uniformly distributed

Hash function

In practice, many different hashing functions are used. One of the most common is the function

$$H(k) = k \mod m$$

where m is the number of available memory locations.

- H is easily evaluated.
- H is not one-to-one but onto on $\mathbb{Z}_m$
- H is not *collision-resistent*.
- The outputs of H are usually uniformly distributed on $\mathbb{Z}_m$.

- Randomly chosen numbers are often needed for computer simulations.
- Different methods have been devised for generating numbers that have properties of randomly chosen numbers.
- Numbers generated by systematic methods are not truly random, they are called pseudorandom numbers.

Linear congruential method

Generate pseudorandom integral numbers in $[0, m]$

- Choose four integers: the modulus m, multiplier a, increment c,and seed x_0, with $2 \leq a < m, 0 \leq c < m$, and $0 \leq x_0 < m$.
- Generate a sequence of pseudorandom numbers x_n, with $0 \leq x_n < m$ for all n, by successively using the recursively defined function

$$x_{n+1} = (ax_n + c) \mod m$$

Generate pseudomrandom numbers in $[0, 1]$

- Generate a sequence of pseudorandom numbers x_n, with $0 \leq x_n < m$ for all n, by Linear congruential method with modulus m.
- Output numbers $y_n = x_n/m$.

Example

Find the sequence of pseudorandom numbers generated by the linear congruential method with

$$m = 11, \quad a = 2, \quad c = 3, \quad x_0 = 1.$$

Solution. We use the recursion

$$x_{n+1} = (2x_n + 3) \pmod{11}.$$

$$\begin{aligned}
x_0 &= 1, \\
x_1 &= (2 \cdot 1 + 3) \bmod 11 = 5, \\
x_2 &= (2 \cdot 5 + 3) \bmod 11 = 13 \bmod 11 = 2, \\
x_3 &= (2 \cdot 2 + 3) \bmod 11 = 7, \\
x_4 &= (2 \cdot 7 + 3) \bmod 11 = 17 \bmod 11 = 6, \\
x_5 &= (2 \cdot 6 + 3) \bmod 11 = 15 \bmod 11 = 4, \\
x_6 &= (2 \cdot 4 + 3) \bmod 11 = 11 \bmod 11 = 0
\end{aligned}$$

$$
\begin{aligned}
x_7 &= (2 \cdot 0 + 3) \bmod 11 = 3, \\
x_8 &= (2 \cdot 3 + 3) \bmod 11 = 9, \\
x_9 &= (2 \cdot 9 + 3) \bmod 11 = 21 \bmod 11 = 10, \\
x_{10} &= (2 \cdot 10 + 3) \bmod 11 = 23 \bmod 11 = 1.
\end{aligned}
$$

Because $x_{10} = x_0$ and each term depends only on the previous term, the sequence is

$$1, 5, 2, 7, 6, 4, 0, 3, 9, 10, 1, \ldots$$

This sequence contains 10 different numbers before repeating.

Popular parameter choice for LCG

- Java's random:

$$m = 2^{48}, a = 25214903917, c = 11.$$

- Park-Miller random generator, known as MINSTD, used in C++11

$$m = 2^{31} - 1 = 2147483647(\text{prime}), a = 7^5 = 16807, c = 0$$

Caesar's *encryption*

One of the earliest known uses of cryptography was by Julius Caesar. He made messages secret by shifting each letter three letters forward in the alphabet.
To define the Caesar's *encryption* mathematically, we represent each English letter by an integer from 0 to 25 :

$$\begin{array}{ccccccccccccc} A & B & C & D & E & F & G & H & I & J & K & L & M \\ \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow \\ 0 & 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 & 11 & 12 \end{array}$$

$$\begin{array}{ccccccccccccc} N & O & P & Q & R & S & T & U & V & W & X & Y & Z \\ \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow \\ 13 & 14 & 15 & 16 & 17 & 18 & 19 & 20 & 21 & 22 & 23 & 24 & 25 \end{array}$$

Caesar's encryption

- **Encryption:** The letter represented by p in $\mathbb{Z}_{26}$ is encrypted into the letter represented by $c = \text{Enc}(p)$ in $\mathbb{Z}_{26}$ such that

$$\textbf{Enc}(p) = (p + 3) \bmod 26$$

- **Decryption:** To decrypt the ciphertext c, compute

$$\textbf{Dec}(c) = c - 3 \bmod 26$$

Caesar cipher example (shift +3)

The message "MEET YOU IN THE PARK" is encrypted as

M	E	E	T	Y	O	U	I	N
12	4	4	19	24	14	20	8	13
↓	↓	↓	↓	↓	↓	↓	↓	↓
15	7	7	22	1	17	23	11	16

T	H	E	P	A	R	K
19	7	4	15	0	17	10
↓	↓	↓	↓	↓	↓	↓
22	10	7	18	3	20	13

P H H W B R X L Q W K H S D U N

Encryption: MEET $\rightarrow$ PHHW (shift $+3$)

We use $\text{Enc}(p) = (p+3) \bmod 26$. Map A$\rightarrow$0, B$\rightarrow$1, ..., Z$\rightarrow$25.

M	E	E	T
12	4	4	19
$\downarrow$	$\downarrow$	$\downarrow$	$\downarrow$
15	7	7	22

P H H W

Decryption: PHHW $\rightarrow$ MEET (shift -3)

We use $\text{Dec}(c) = (c - 3) \bmod 26$. Same letter$\leftrightarrow$number mapping.

P	H	H	W
15	7	7	22
$\downarrow$	$\downarrow$	$\downarrow$	$\downarrow$
12	4	4	19

M E E T

Shift cipher

- Choose a **secret number** t in $\mathbb{Z}_{26}$
- **Encryption:**

$$Enc(p) = (p + t) \pmod{26}$$

- **Decryption:**

$$Dec(c) = (c - t) \pmod{26}$$

Note that in Shift Cipher, **Enc**(p) is a bijection and **Dec** is the inverse function of **Enc**.

We can improve the encryption using the function

$$\mathbf{Enc}(p) = (ax + b) \bmod 26$$

where a and b are secret constants chosen so that this function is a bijection.

Improved Shift Cipher

- **Secret key:** (a, b) where $\gcd(a, 26) = 1$.
- **Encryption** (p : letter):

$$\mathrm{Enc}(p) = a \cdot p + b \pmod{26}$$

- **Decryption** (c):

$$\mathrm{Dec}(c) = a' \cdot (c - b) \pmod{26}$$

where a' is the inverse of $a \pmod{26}$.

Exercise

Set $\mathbb{Z}_{26} = \{0, 1, 2, \ldots, 25\}$

a) List all the positive integers in $\mathbb{Z}_{26}$ that are invertible modulo 26.

b) Let a and b be integers. Consider the function

$$f : \mathbb{Z}_{26} \longrightarrow \mathbb{Z}_{26}$$

defined by

$$f(x) = ax + b \pmod{26}.$$

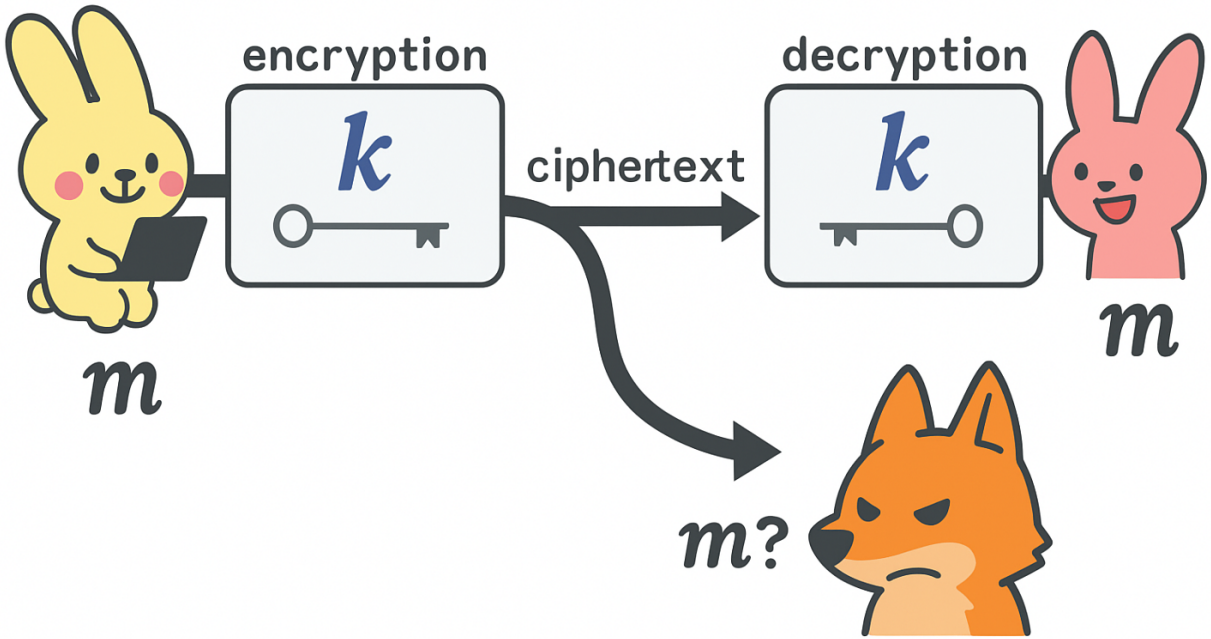
i) Prove that $f(x) = 3x$ is bijective and find the inverse function.
ii) Prove that $f(x) = 3x + b$ is bijective for any b.
iii) Find all the values of a for which the function f is bijective.

Is Shift Cipher secure ?

Try to decrypt the following ciphertext that was generated using the shift cipher without knowing k
OVDTHUFWVZZPISLRLFZHYLAOLYL
Hint: there are only 26 possible keys. Try every possible keys and check which key results in a plaintext that make sense.

Private key encryption

In the setting of private-key encryption, two parties share a key and use this key when they want to communicate secretly. One party can send a message, or plaintext, to the other by using the shared key to encrypt the message and thus obtain a ciphertext that is transmitted to the receiver. The receiver uses the same key to decrypt the ciphertext and recover the original message.

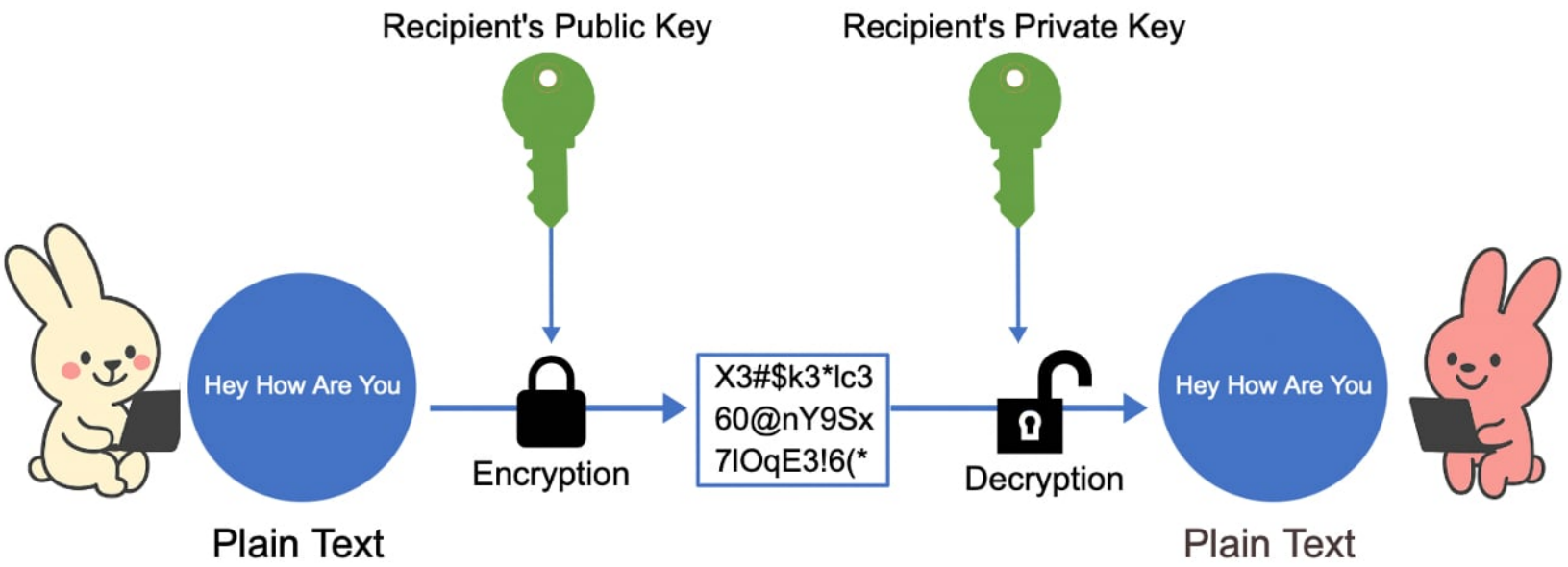


Public key encryption

A public key encryption scheme works as follows:

1. **Key generation:**
   - Bob generates his own key pair **(public key, private key)**.
   - Bob keeps the private key secret and shares the public key openly.
2. **Encryption (done by Alice):**
   - Alice wants to send Bob a confidential message m.
   - She uses Bob's public key to encrypt m, producing a ciphertext c.
   - $c \leftarrow \text{Enc}(\text{Bob's public key}, m)$.
   - She sends the ciphertext c to Bob.
3. **Decryption (done by Bob):**
   - Bob receives c.
   - He uses his private key to decrypt c, recovering the original message m.
   - $m = \text{Dec}(\text{Bob's private key}, c)$.

Public key encryption scheme



A public key encryption scheme consists of three algorithms:

1. **KeyGen**(λ): takes as input a security parameter λ. It outputs a key pair consisting of a public key and a secret key (pk, sk) for each user.
2. **Enc**(pk, m): takes as input the recipient's public key pk and a message m. It outputs a ciphertext c.
3. **Dec**(sk, c): takes as input the recipient's secret key sk and a ciphertext c. It outputs the original message m or a special error symbol $\perp$.

RSA public key encryption scheme

1. **KeyGen()**:
   - Choose two large distinct primes p, q.
   - Compute modulus $n = pq$.
   - Compute Euler totient $\varphi(n) = (p-1)(q-1)$.
   - Choose public exponent e with
   $$1 < e < \varphi(n), \qquad \gcd(e, \varphi(n)) = 1.$$
   - Compute private exponent d as the modular inverse of e mod $\varphi(n)$:
   $$de \equiv 1 \pmod{\varphi(n)}.$$
   - Output
   $$\textbf{Public key} = (n, e), \qquad \textbf{Private key} = d.$$
2. **Enc($pk = (n, e), m$):** compute the ciphertext
   $$c = m^e \pmod{n}.$$
3. **Dec($pk = (n, e), m$):** Recover message m
   $$m = c^d \pmod{n}.$$

RSA: Correctness (Why decryption works)

Since $de \equiv 1 \pmod{\varphi(n)}$, there exists k with $de = 1 + k\,\varphi(n)$. For m coprime to n,

$$m^{de} = m^{1+k\varphi(n)} = m \cdot \left(m^{\varphi(n)}\right)^k \equiv m \cdot 1^k \equiv m \pmod{n}$$

by Euler's theorem $\left(m^{\varphi(n)} \equiv 1 \pmod{n}\right)$.

Security Note (Textbook vs. Real RSA)

- **Textbook RSA** (deterministic $c = m^e \bmod n$) is *not semantically secure* and is *malleable*. It should not be used directly.
- Keys must be generated with strong randomness, and primes p, q must be large and secret.

Real world RSA

Primitives for Real world RSA encryption scheme

1. **Private key / Symmetric encryption scheme**
   - $\mathsf{KeyGen}() \rightarrow k$
   - $E_S = \mathsf{Enc}(k, m) \rightarrow c$
   - $D_S = \mathsf{Dec}(k, c) \rightarrow m$
2. **Hash function** $H(x) \rightarrow k$
3. **Textbook RSA encryption and decryption**
   - $\mathsf{RSA.Enc}(x) = x^e$
   - $\mathsf{RSA.Dec}(y) = y^d$

Real World RSA Encryption Scheme

1. **Key generation:** like textbook RSA KeyGen
   - RSA modulus $N = pq$, public exponent e, private exponent d.
   - Public key: (N, e), secret key: d.
2. **Encryption** (pk, m)**:**
   - Pick random $x \in \{0, 1, \ldots, N-1\}$.
   - Derive $k = H(x)$ (a symmetric key).
   - Compute $c = E_S(k, m)$ (symmetric encryption of m under k).
   - Encrypt x with RSA: $y = x^e \pmod{N}$.
   - Ciphertext is $C = (c, y)$.
3. **Decryption** (sk, C)**:**
   - Recover $x = y^d \pmod{N}$.
   - Recompute $k = H(x)$.
   - Decrypt: $m = D_S(k, c)$.